AMD Physical AI
Technical Report

AMD AI DevMaster Radeon GPU Hackathon
Track 3: Physical AI Challenge

Team: Pioneer
Members: Sa Su, Zecheng Li
Date: 2026-08-06
Submission: Track 3, Pioneer, Soda Can Pick-and-Place (+ Stretch Cleanup)

ROCm 7.2 | AMD Radeon | Genesis + MuJoCo | LeRobot v3

1 Application Definition

This project builds a complete robotic manipulation pipeline on AMD Radeon GPUs, covering simulation, data generation, policy training, and closed-loop evaluation. The primary task is a Franka Panda arm picking a soda can from a table and placing it into a target bin.

A secondary extension explores a mobile manipulator in a living-room cleanup scenario, demonstrating the same pipeline's generality to navigation plus grasping.

2 System Architecture

The system follows a three-stage sensor-to-control pipeline running on a single AMD Radeon GPU node.

Stage	Model	Input	Output
Perception	YOLOE-26S	RGB-D frames	Object boxes + masks
Navigation	NoMaD	4-frame nav images	8 waypoints (x, y)
Manipulation	SmolVLA	Head + Wrist + Mask + 6-DOF state	6-DOF action block

Perception detects objects in RGB-D and back-projects to 3D using median depth. Navigation takes four consecutive frames and outputs waypoints for the mobile base. Manipulation takes head camera, wrist camera, and a target-mask view, then outputs lift, arm extension, wrist yaw/pitch/roll, and gripper commands.

All three models run in real time on a single AMD Radeon GPU node, synchronized every control cycle.

3 Simulation Environment

3.1 Franka Soda-Can Picking

Built on the Genesis physics engine with a Franka Panda 7-DOF arm. The target object is a red soda can modeled as a rigid cylinder with realistic mass and friction. A scripted expert policy computes pick-and-place trajectories using inverse kinematics, which are recorded as LeRobot v3 datasets.

Domain randomization is applied to object color, table texture, lighting, and camera FOV to improve policy robustness.

3.2 Stretch Living-Room Cleanup

Built on MuJoCo with a Hello Robot Stretch 3 mobile manipulator. The scene contains a living room with floor, furniture, and four pickable objects: can, bottle, carton, and cup. A browser-based teleoperation interface allows human operators to control the arm directly via mouse and keyboard, recording actions as executioner increments.

Key design choices: no inverse kinematics, no equality or weld constraints, no object ground-truth access during inference. An integrity monitor rejects runs with non-physical displacements or constraint violations.

4 Dataset

Dataset	Episodes	Frames	Objects	Annotations
Franka soda_pick	50	~15,000	Soda can (red)	LeRobot v3
Stretch human_v1	50	~30,000	Can/Bottle/Carton/Cup	MuJoCo seg
Livingroom_v2	2,300 scenes	~115,000	4 classes	Auto-labeled

The perception dataset livingroom_v2 contains 100,000 training and 15,000 held-out images with 181,681 bounding-box annotations across four classes. Held-out scenes use disjoint seeds for room size, furniture layout, lighting, and floor color, providing a true generalization test.

5 AMD Radeon GPU and ROCm

All experiments run on a single node with four AMD Radeon GPUs (gfx1201, RDNA4, 30.5 GB each), driven by ROCm 7.2.3 and PyTorch 2.10.0+rocm. The full software stack is: AMD GPU + amdgpu driver, ROCm + HIP + MIOpen, PyTorch + AOTriton, Genesis + Taichi for simulation, LeRobot + Transformers for policy.

Model	p50 latency	p95 latency	Device
YOLOE-26S	5.20 ms	7.39 ms	GPU 0
NoMaD	52.10 ms	68.30 ms	GPU 3
SmolVLA	54.80 ms	69.01 ms	GPU 3

All three models exceed 10 Hz. Raw tensors, input hashes, weight SHAs, and HIP device IDs are logged to outputs/realtime/ for full traceability.

Data generation reaches 655 images per second using 96 CPU processes, saturating all 192 threads of a Threadripper PRO 9995WX.

6 Training

SmolVLA is fine-tuned on the recorded LeRobot datasets using the standard LeRobot training script. Training runs on a single AMD Radeon GPU with batch size 4 to 8 depending on model size. Checkpoints are saved every 500 steps and evaluated on a held-out set.

NoMaD wheel dynamics is trained on navigation trajectories generated by a scripted exploration policy. YOLOE is trained on livingroom_v2 with 12 epochs at 640x640 resolution.

7 Evaluation and Results

Test	Result	Notes
YOLOE held-out recall	Measured	Livingroom_v2 held-out set
NoMaD closed-loop	10/10 collision-free	Short-range navigation
SmolVLA grasping	Evaluated	Recorded episodes + closed-loop
Integrity tests	5/5 pass	No teleport, no weld, no cheat
Real-time sync	All >10 Hz	p95 logged per model

A demo video shows the full pipeline: command-line launch, ROCm GPU utilization on screen, simulation rendering, and the robot completing pick-and-place. The video is 3 to 5 minutes long.

8 Innovation and Contributions

1. End-to-end pipeline on AMD Radeon: simulation, data, training, and inference all on ROCm.
2. Integrity monitoring that rejects non-physical runs, ensuring all results come from valid physics.
3. Large-scale auto-labeled perception dataset with disjoint train/held-out splits for honest generalization testing.
4. Real-time synchronized three-model inference exceeding 10 Hz on a single ROCm node.
5. Open-source contributions back to the upstream Franka demo repository.

9 Deliverables

Item	Format	Location
Technical Report	PDF	This document
Source code	Git repo	fork of Radeon-hackathon-2026-07
README	Markdown	track3_soda_pick/README.md
Demo video	MP4, 3-5 min	track3_soda_pick/media/

Item	Format	Location
Dataset	LeRobot v3	datasets/soda_pick/
Trained weights	Checkpoint	outputs/smolvla_soda_pick/
Dockerfile	Text	track3_soda_pick/Dockerfile

10 Reproducibility

1. Install dependencies: pip install -e . and pip install lerobot[training,smolvla]==0.6.0
2. Verify ROCm: python check_rocm.py
3. Build scene: python franka_fruit_pick/build_scene.py --steps 50 --save-frames
4. Record data: python franka_fruit_pick/record_dataset.py --episodes 50 --repo-id local/soda_pick
5. Train: python franka_fruit_pick/train_policy.py smolvla --repo-id local/soda_pick
6. Evaluate: python franka_fruit_pick/eval_policy.py --episodes 20 --save-video
7. Realtime stack: python -m realtime_cleanup.smoke --perception-device 0 --navigation-device 3 --manipulation-device 3

11 Team Members and Contributions

Member	Role	Contributions
Sa Su	Simulation + Training	Genesis scene, LeRobot data pipeline, SmolVLA fine-tuning, ROCm optimization
Zecheng Li	Perception + Navigation + Integration	YOLOE training, NoMaD integration, real-time sync, integrity tests, code integration, README, demo video, technical report

12 Conclusion

This work builds and validates a complete physical AI pipeline on AMD Radeon GPU + ROCm platform. The three-model real-time inference exceeds 10 Hz, data generation and training are fully on ROCm, and the integrity monitoring system ensures all results come from valid physics.

The Franka soda-can pick-and-place task demonstrates the core pipeline end-to-end. The Stretch living-room cleanup extension shows the same approach generalizes to mobile manipulation with navigation.